Quantification: 872_wt_hp_z06
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PRE-VALIDATION: no number here has been checked against manual (ImageJ) counts.
PIXEL UNITS: lengths px, areas px^2, densities px^-1. The images carry no micron calibration.
Statistics are over interior lacunae only (those not touching the frame edge).

Per image
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measure	value
lacuna count	15
interior lacuna count	14
frame-edge lacuna count	1

Interior lacunae only: mean, median and sample SD (n = 14)
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measure	kind	mean	median	sd	units
area_px2	lacuna shape	1465.93	1361.5	676.142	px^2
major_axis_length_px	lacuna shape	82.1504	81.9504	23.5356	px
minor_axis_length_px	lacuna shape	25.1534	22.6921	7.9468	px
aspect_ratio	lacuna shape	3.5417	3.5122	1.5974	unitless
eccentricity	lacuna shape	0.9308	0.9586	0.0723	unitless
solidity	lacuna shape	0.809	0.8566	0.109	unitless
roots_count	headline	10.2857	9.5	4.7625	count
ring_length_r30_px	headline	435.071	387.5	130.829	px
ring_length_r60_px	reported	1077.14	1008.5	365.728	px
owned_length_px	ownership-dependent	2488.62	2515.11	1291.31	px
edge_count	ownership-dependent	156.143	154	87.5002	count
mean_edge_length_px	ownership-dependent	16.1872	15.9293	1.2839	px
perimeter_px	normalised	212.522	216.238	57.6792	px
ring_area_r30_px2	normalised	8201.36	8560.5	1218.55	px^2
ring_area_r60_px2	normalised	20241.2	21261.5	2779.04	px^2
in_frame_fraction_r30	normalised	0.9924	1	0.0286	unitless
in_frame_fraction_r60	normalised	0.9864	1	0.0509	unitless
ring_density_r30	normalised	0.0531443	0.0515273	0.0136295	px^-1
ring_density_r60	normalised	0.0530231	0.0542399	0.0162992	px^-1
roots_per_100px_perimeter	normalised	4.8883	5.2241	1.9998	per 100 px
ring_attached_length_r30_px	network v2	407.929	361	129.367	px
ring_attached_length_r60_px	network v2	949.571	839	395.541	px
ring_length_w_r30_px	network v2	494.423	442.798	151.357	px
ring_length_w_r60_px	network v2	1225.64	1138.94	417.736	px
sholl_crossings_r10	network v2	9.4286	9	3.5239	count
sholl_crossings_r20	network v2	12.2143	12	4.2999	count
sholl_crossings_r30	network v2	14	14.5	4.6244	count
loop_count	loops, report only	21.4286	19	17.1451	count
loop_edge_share	loops, report only	0.4535	0.4525	0.1699	share
ring_width_mean_r30_px	width	7.4094	7.511	0.5525	px

Per field
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measure	value	units
analysed_area_px2	1.02738e+06	px^2
total_skeleton_length_px	39179	px
canalicular_length_density_per_px	0.0381347	px^-1
junction_count	1320	count
junction_density_per_px2	0.00128482	px^-2
lacuna_count	15	count

lacunae_per_px2	1.46e-05	px^-2
skeleton_component_count	153	count
mean_component_length_px	256.072	px
median_component_length_px	15	px
field_length_density_w_per_px	0.0432452	px^-1
field_density_without_flagged_per_px	0.0495713	px^-1
field_density_in_roi_per_px		px^-1
unattached_length_px	23	px
unattached_component_count	2	count
canalicular_length_density_attached_per_px	0.0381347	px^-1
field_length_density_with_unattached_per_px	0.0381571	px^-1
loop_count	391	count
loop_edge_share	0.5474	share
edge_count	2627	count

Canalicular width, the new measure

measure	value	units
width_mean_px	7.6414	px
width_median_px	8	px
width_p10_px	4	px
width_p90_px	10.7703	px
width_pixels_used	37640	count

Canalicular width: 2 x the Euclidean distance transform of the canalicular mask at each skeleton pixel. Measured only on skeleton pixels on the canalicular mask: the lacuna buffer and the dilated vascular mask are excluded. The threads are about 2 to 6 px wide, so the width is close to the pixel grid and quantised in steps of about 1 px, it moves with the threshold (a lower cut widens every thread), and it has not been validated against any manual measurement.

Lacuna shape

lacuna_id	area_px2	major_axis_length_px	minor_axis_length_px	aspect_ratio	eccentricity	solidity	orientation_rad	centroid_row_px	centroid_col_px	on_border
1	1057	102.462	18.2811	5.6048	0.984	0.7152	-1.5145	124.814	413.139	False
2	1545	102.844	28.21	3.6457	0.9616	0.5974	-1.5363	130.782	502.108	False
3	2178	97.2133	31.5882	3.0775	0.9457	0.7666	-1.4507	132.953	694.614	False
4	1024	110.295	13.9978	7.8795	0.9919	0.6469	1.515	141.115	876.421	False
5	1532	90.3989	22.6854	3.9849	0.968	0.8855	-1.4786	133.692	263.431	False
6	1458	82.9994	23.987	3.4602	0.9573	0.8423	-1.4301	272.217	418.857	False
7	2267	64.9571	46.3919	1.4002	0.6999	0.896	-1.1815	380.946	282.045	False
8	3165	121.397	33.5624	3.6171	0.961	0.8709	-1.4322	386.945	67.1896	False
9	743	47.9512	20.7766	2.3079	0.9013	0.8888	-1.5064	382.218	839.734	False
10	712	46.6981	20.2555	2.3054	0.901	0.8856	-1.5028	499.643	510.035	False
11	1363	80.3055	21.9882	3.6522	0.9618	0.8961	-1.3459	561.869	697.237	False
12	669	39.9711	24.8769	1.6068	0.7827	0.7707	-1.1357	585.831	14.0269	True
13	1360	67.805	25.7054	2.6378	0.9254	0.9484	-1.4973	614.482	323.551	False
14	846	53.8777	22.02	2.4468	0.9127	0.8004	1.5387	642.791	211.617	False
15	1273	80.9015	22.6988	3.5641	0.9598	0.6866	-1.5697	768.217	476.052	False

Network per lacuna

lacuna_id	roots_count	ring_length_r30_px	ring_length_r60_px	owned_length_px	edge_count	mean_edge_length_px	ring_width_mean_r30_px
1	6	244	442	457.191	28	16.3282	7.5301
2	8	373	713	991.607	67	14.8001	7.887
3	10	465	884	1747.05	122	14.3201	7.8301
4	12	379	965	2499.16	151	16.5507	7.7983

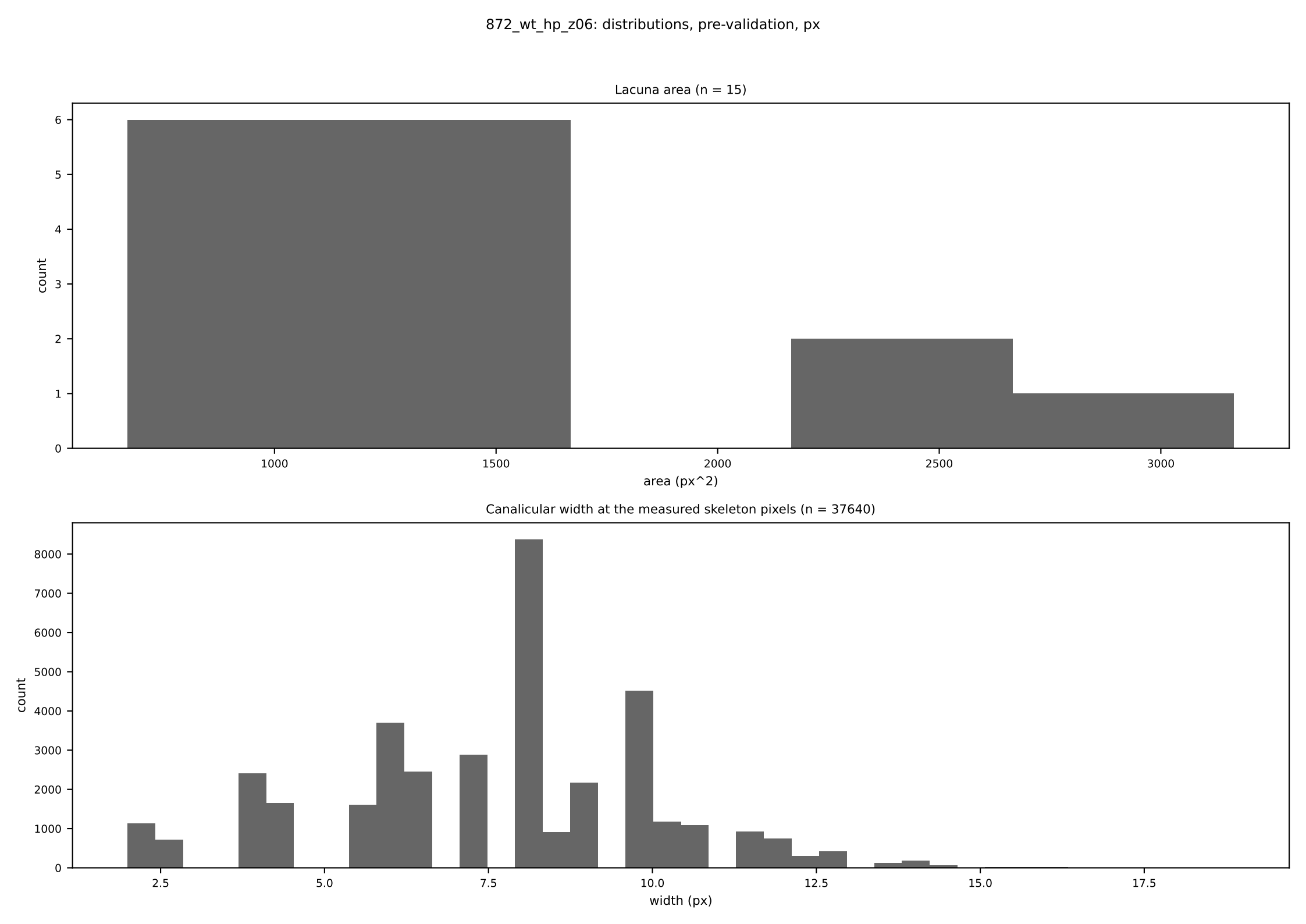
5	12	349	696	2531.05	159	15.9186	6.7785
6	20	568	1570	3235.21	206	15.7049	7.7049
7	13	578	1524	2231.61	140	15.9401	8.0859
8	18	741	1762	5347.95	367	14.5721	7.3404
9	8	370	1051	3042.28	173	17.5854	7.2474
10	4	309	986	1540.44	89	17.3083	6.4548
11	12	476	1204	2866.19	157	18.256	6.958
12	8	353	919	2462.03	161	15.2921	7.9105
13	3	396	1031	1189.72	65	18.3034	6.5111
14	9	509	1305	4099.43	266	15.4114	8.1131
15	9	334	947	3061.83	196	15.6216	7.4919

Normalised per lacuna

lacuna_id	perimeter_px	ring_area_r30_px2	ring_area_r60_px2	in_frame_fraction_r30	in_frame_fraction_r60	ring_density_r30	ring_density_r60	roots_per_100px_perimeter
1	231.817	7355	14578	1	1	0.0331747	0.0303197	2.5882
2	285.137	8557	19005	1	1	0.04359	0.0375164	2.8057
3	292.451	10015	23365	1	1	0.0464304	0.0378344	3.4194
4	241.681	9364	22992	1	1	0.0404742	0.0419711	4.9652
5	218.853	8669	20339	1	1	0.0402584	0.03422	5.4831
6	215.095	8592	22200	1	1	0.066108	0.0707207	9.2982
7	212.409	8564	22184	1	1	0.0674918	0.0686982	6.1203
8	301.179	9919	22490	0.893	0.8096	0.0747051	0.0783459	5.9765
9	123.154	6283	18209	1	1	0.0588891	0.0577187	6.4959
10	119.983	6180	18000	1	1	0.05	0.0547778	3.3338
11	196.024	8388	22420	1	1	0.0567477	0.0537021	6.1217
12	129.29	4152	11045	0.6578	0.6072	0.0850193	0.0832051	6.1876
13	167.539	7464	18130	1	1	0.0530547	0.0568671	1.7906
14	152.61	6844	16739	1	1	0.0743717	0.0779616	5.8974
15	217.38	8625	22726	1	1	0.0387246	0.0416703	4.1402

Network v2 per lacuna

lacuna_id	ring_attached_length_r30_px	ring_attached_length_r60_px	ring_length_w_r30_px	ring_length_w_r60_px	sholl_crossings_r10	sholl_crossings_r20	sholl_crossings_r30
1	236	413	279.551	503.576	6	8	5
2	370	667	444.132	831.139	6	8	10
3	456	850	528.262	994.654	10	12	10
4	352	965	420.037	1083.83	8	12	14
5	347	602	401.635	796.713	7	12	10
6	562	1549	639.203	1779.84	13	18	16
7	531	1420	661.973	1752.52	11	16	21
8	702	1735	850.754	2008.66	19	22	21
9	330	745	416.806	1219.37	10	9	16
10	244	584	340.208	1117.06	6	7	15
11	436	849	543.961	1362.57	11	15	19
12	347	887	409.806	1051.35	6	11	10
13	347	798	441.463	1160.82	8	10	12
14	484	1288	582.203	1494.06	10	13	16
15	314	829	371.735	1054.18	7	9	11


872_wt_hp_z06: distributions, pre-validation, px
Lacuna area (n = 15)
count
area (px^2)
0
1
2
3
4
5
6
1000
1500
2000
2500
3000
Canalicular width at the measured skeleton pixels (n = 37640)
count
width (px)
0
1000
2000
3000
4000
5000
6000
7000
8000
2.5
5.0
7.5
10.0
12.5
15.0
17.5